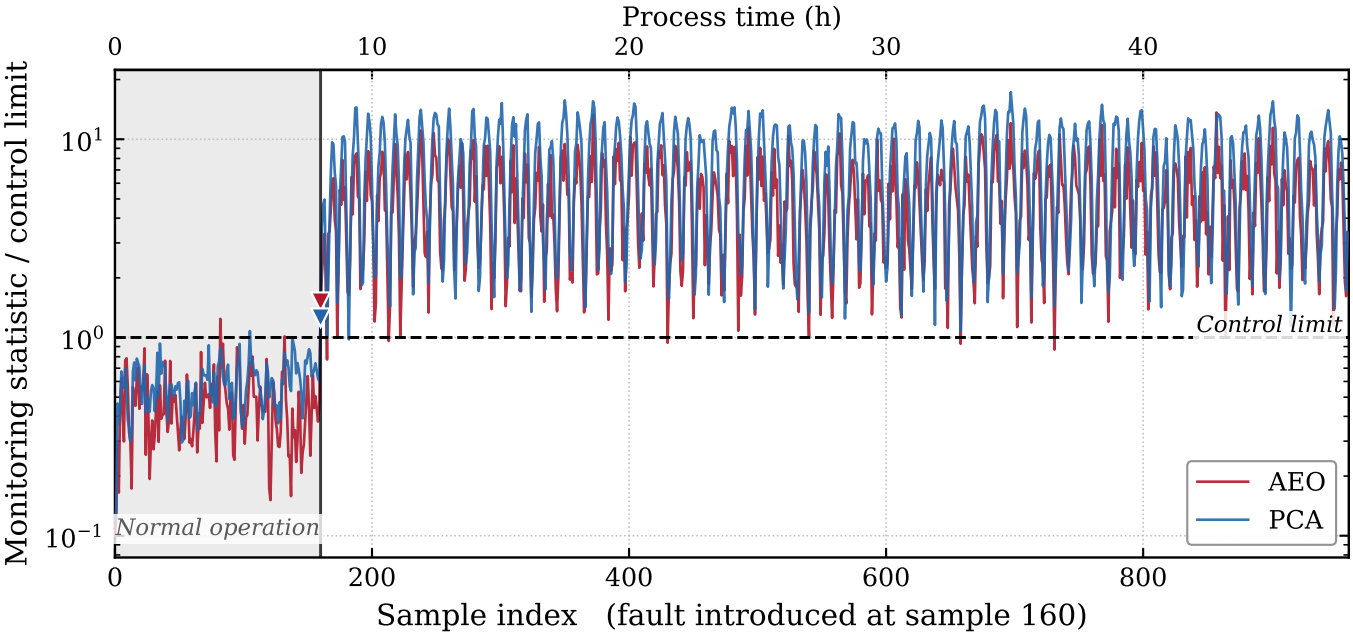
Fault Detection Performance for IDV(14)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	2.0 samples (6 min)	MDR 0.014	detected in 7/7 runs
PCA	delay	0.4 samples (1 min)	MDR 0.002	detected in 7/7 runs

PCA detects 1.6 samples (5 min) earlier; PCA missed-detection rate lower by 87%